Assignment 1

Exploratory Data Analytics

KQC 7076
Semester 2, 2025/2026

Instructions

- This assignment aims to help you apply Exploratory Data Analysis (EDA) and statistical hypothesis testing to uncover insights from real-world datasets using Python.
- You will demonstrate your ability to explore, clean, analyze, and interpret data scientifically.
- You will work in pairs (2 students per group), and each pair will work on a unique dataset/theme assigned.
- You must complete and submit both a Jupyter Notebook and a brief report (~10 pages).

Themes

Max 5 groups per theme

1. Environmental
2. Smart Manufacturing
3. Intelligent Transportation System
4. Smart City
5. Smart Energy
6. Smart Agriculture
7. Autonomous Driving
8. Smart Hospital

Report Format (max 10 pages)

- Cover Page
- Introduction
- Description of Datasets
- Exploratory Data Analysis (EDA)
- Statistical Analysis
- Conclusion
- References

Rubrics

Criteria	5	4-3	2-1
Introduction (10%)	Clearly defines problem statement and objectives; provides solid context and relevance of dataset; research questions are precise and logical.	Basic description of data and objectives; research questions are vague.	No clear objective; dataset or problem poorly explained.
Data Understanding & Preparation (15%)	Comprehensive data description; demonstrates cleaning, handling of missing values, and preprocessing; summary stats and correlation analysis done correctly.	Limited cleaning and description; basic understanding shown.	No evidence of data understanding or cleaning; poor code documentation.
Exploratory Data Analysis (EDA) (25%)	Excellent use of visualizations; insightful interpretation of trends, patterns, and anomalies; plots are labeled and well-presented.	Few or repetitive plots; interpretation is minimal or generic.	Inadequate or incorrect visualizations; little to no analysis or discussion.

Rubrics

Criteria	5	4-3	2-1
Statistical Analysis (25%)	Correct selection and execution of tests; assumptions checked; results interpreted accurately with relevant post-hoc tests; strong linkage to research question.	Tests used but with conceptual or interpretation errors; results not well-linked to questions.	Wrong or unjustified test selection; results misinterpreted or missing; poor statistical reasoning.
Discussion (15%)	Integrates EDA and statistical findings logically; demonstrates critical thinking; acknowledges limitations and context.	Basic summary without connecting EDA and statistical insights; little reasoning.	Superficial or missing discussion; no critical thinking evident.
Conclusion (5%)	Clear and concise summary of findings; directly answers research objectives; practical relevance highlighted.	Brief and generic conclusion; weak linkage to earlier analysis.	No clear conclusion; poorly written or irrelevant.

Rubrics

Criteria	5	4-3	2-1
Report Structure, Code Quality, and References (5%)	Professional presentation; logical organization; Python code is clean, commented, and reproducible; all references properly cited.	Adequate structure; inconsistent formatting or incomplete code documentation.	Disorganized report; poor writing; missing references or unreadable code.